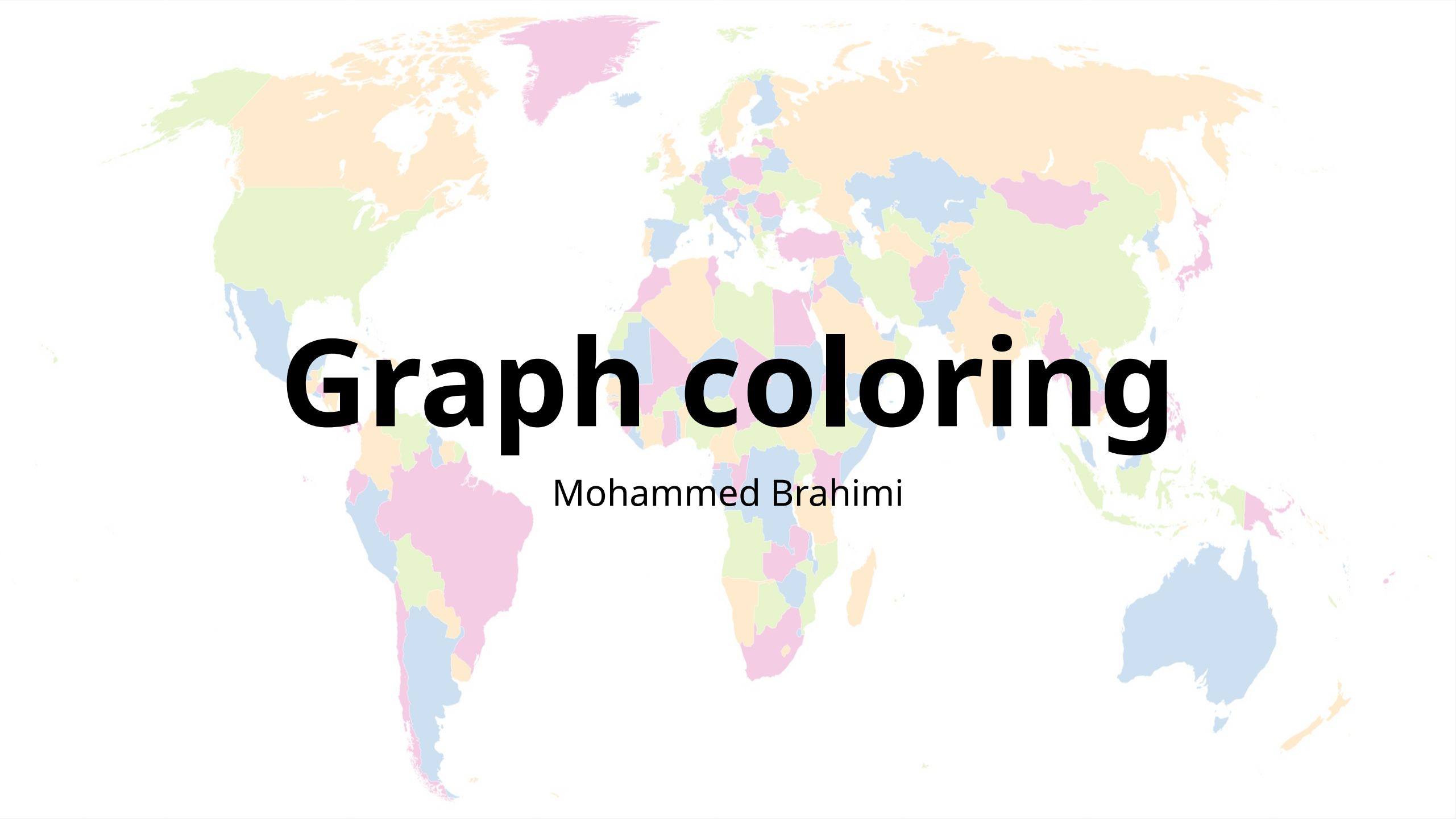




Graph coloring

Mohammed Brahimi

Module distribution concerns raised by students

- Issues
 - Multiple difficult modules scheduled in the same term
 - Unbalanced distribution of workload between terms
- Goal
 - Achieving a balanced load distribution between terms
- Approach:
 - Represent modules as vertices
 - High-load modules are linked by edges
 - Assign colors to terms
 - Ensure that difficult and high-load modules are not scheduled in the same term.



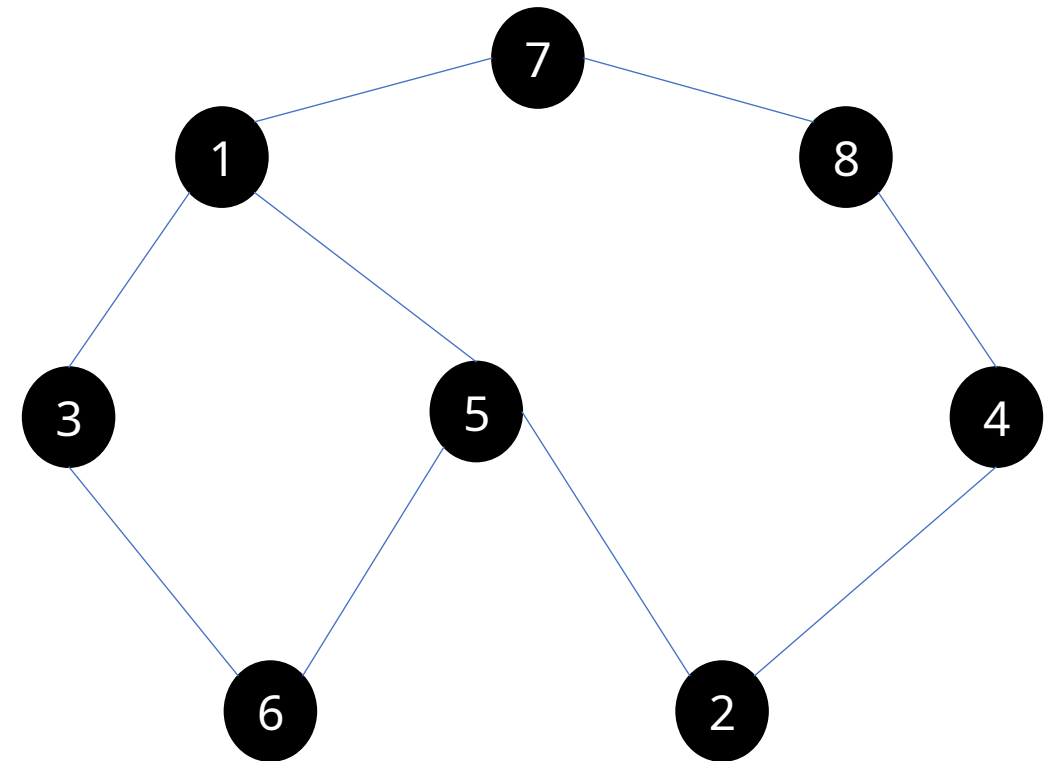
Find optimal module-term assignment to minimize load imbalance and avoid clustering high-load modules.

How to avoid challenging modules in the same term?

		1	2	3	4	5	6	7	8
Analysis 3	1			X		X		X	
Data structure	2				X	X			
Mathematical Logic	3	X					X		
Architecture 1	4		X						X
Complexity	5	X	X				X		
Linear Algebra 2	6			X		X			
Probability and Statistics	7	X							X
Databases	8				X			X	

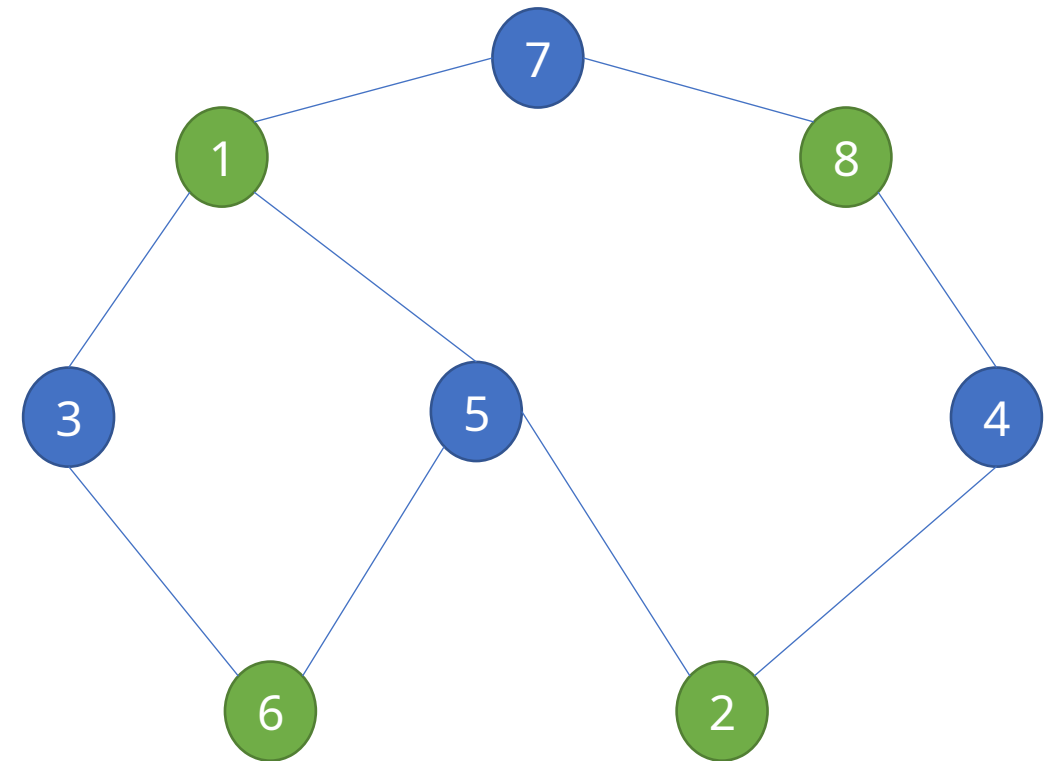
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		1	2	3	4	5	6	7	8
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Complexity	5	X	X				X		
Linear Algebra 2	6			X		X			
Probability and Statistics	7	X							X
Databases	8				X			X	



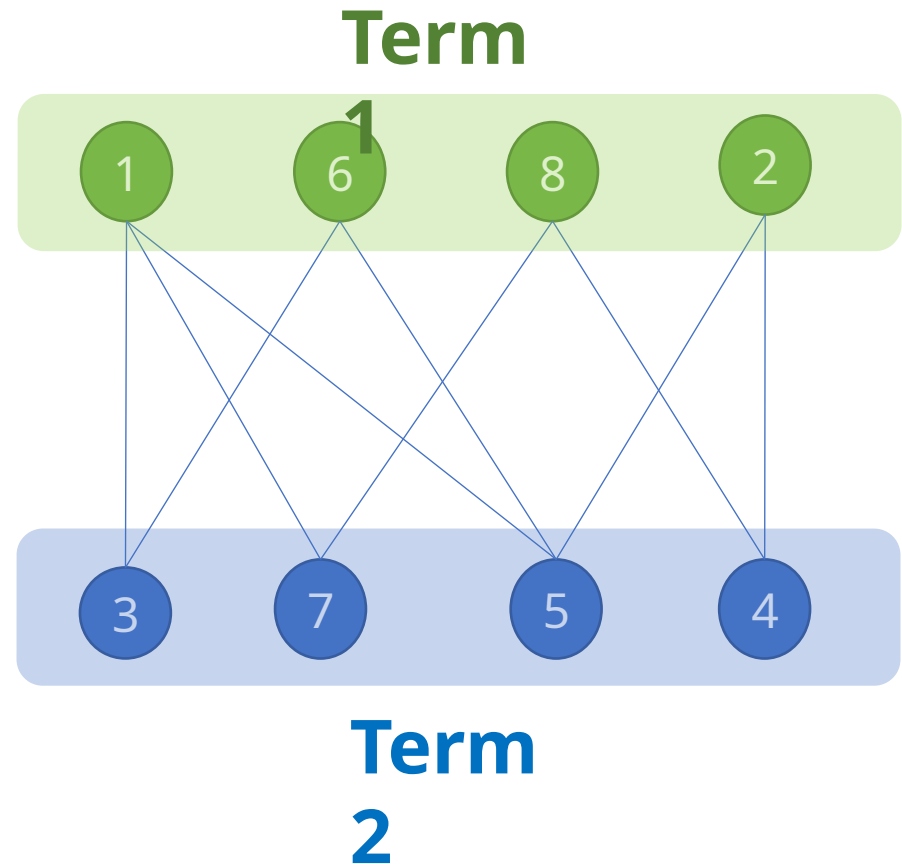
How to avoid challenging modules in the same term?

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Analysis 3	1			X		X		X	
Data structure	2				X	X			
Mathematical Logic	3	X					X		
Architecture 1	4		X						X
Complexity	5	X	X				X		
Linear Algebra 2	6			X		X			
Probability and Statistics	7	X							X
Databases	8				X			X	



How to avoid challenging modules in the same term?

		1	2	3	4	5	6	7	8
Analysis 3	1			X		X		X	
Data structure	2				X	X			
Mathematical Logic	3	X					X		
Architecture 1	4		X						X
Complexity	5	X	X				X		
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Probability and Statistics	7	X							X
Databases	8				X			X	



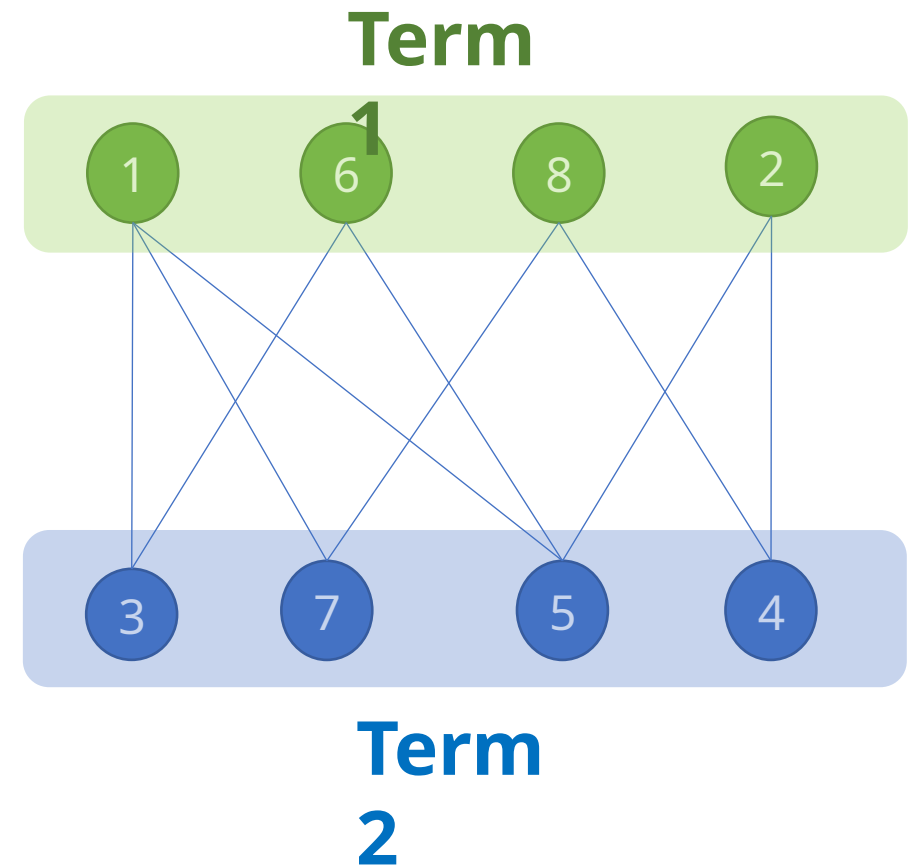
How to avoid challenging modules in the same term?

- **Term 1**

- Data structure
- Linear Algebra 2
- Databases
- Analysis 3

- **Term 2**

- Architecture 1
- Mathematical Logic
- Complexity
- Probability and Statistics



Problem extension

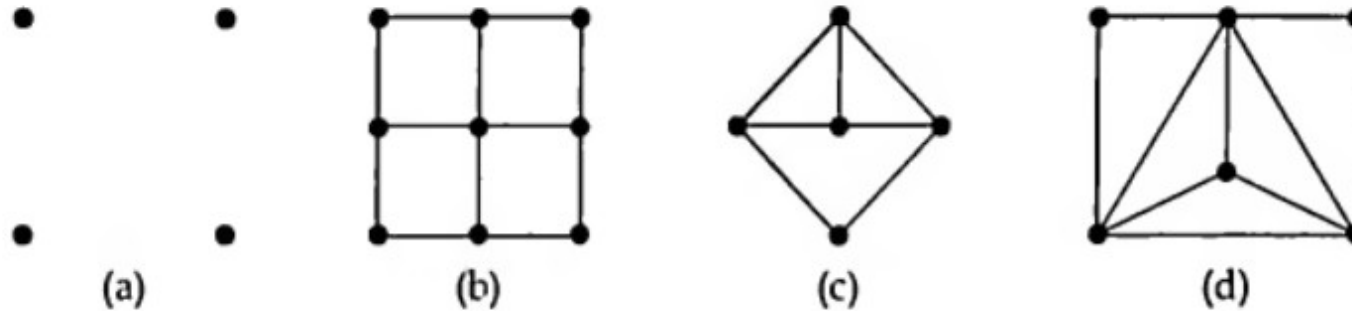
- What if a teacher teach two modules together:
 - Assign a unique color to each teacher.
 - Modules taught by the same teacher are adjacent vertices in the graph.
 - Find a suitable coloring for the graph representing the modules and teachers.
- What about other time table constraints:
 - Sessions represented as vertices.
 - Make two vertices adjacent based on any constraints.
 - Ensure that conflicting sessions, are not scheduled in parallel.

Create a well-structured time table that considers teacher availability, room constraints, and avoids conflicts between modules.

What is graph coloring ?

- In the context of a simple graph G , a **k-coloring** refers to the assignment of at most k colors to its vertices.
- **k-coloring** should color the adjacent vertices in G by different colors.
- If G has a valid **k-coloring**, we say G is **k-colorable**.
- The **chromatic number** of G , denoted by $\chi(G)$, is the smallest number k for which G is k -colorable.
- Reason for restricting coloring to simple graphs:
 - Vertex with loop cannot be assigned a different color from itself.
 - The presence of one or multiple edges between two vertices has a similar effect, requiring them to be colored differently.

What is the **chromatic number** $\chi(G)$?



a) $\chi(G) = 1$

- One color is sufficient
- Absence of edges

b) $\chi(G) = 2$

- +1 color is required (presence of edges)
- It exists 2-coloring.

a) $\chi(G) = 3$

- + 3 colors are required (presence of triangle)
- It exists 3-coloring.

b) $\chi(G) = 4$

- +4 colors are required (presence of K_4)
- It exists 4-coloring.

Subgraphs and the Chromatic Number

- Subgraphs play a crucial role in understanding the chromatic number of a graph.
- If a subgraph H of G has a chromatic number $\chi(H)$, then $\chi(G)$ must also be at least that value.
- The presence of complete subgraphs (cliques) can increase the chromatic number of a graph.

The subgraphs with the highest chromatic numbers can provide insights into the chromatic number of the entire graph.

THEORME 1 (coloring bipartite graph)

A graph is bipartite if and only if it is $\chi(G)=2$

- Direction 1: Bipartite graph $\implies$ Chromatic number is 2.
 - In a bipartite graph, vertices can be split into two sets, A and B, where all edges connect vertices from different sets.
 - Assign **color 1 to set A**
 - Assign color 2 **to set B**.
 - This yields a valid 2-coloring.
 - Thus, the chromatic number of a bipartite graph is 2.
- Direction 2: Chromatic number is 2 $\implies$ Graph is bipartite.
 - If a graph can be colored with only two colors:
 - Put all the vertices of color 1 to the set A
 - Put all the vertices of color 2 to the set B
 - Two vertices from A (or B) are not adjacent because they have the same color.
 - This results in a partition of the graph into sets A and B where the vertices of each set are not adjacent.

What is the chromatic number ?

- Complete graph K_n
 - All the vertices are connected
 - $\chi(G)=n$
- Path graph $P_n (n>1)$
 - Alternate the colors
 - $\chi(G)=2$
- Cycle graph $C_n (n>2)$
 - $\chi(G)=2$, if $n = 2k$
 - $\chi(G)=3$, if $n = 2k+1$
- Wheel graph $W_n (n>1)$
 - $\chi(G) = 4$, if $n = 2k$
 - $\chi(G) = 3$, if $n = 2k+1$

THEORME 2

let G be a simple graph whose maximum vertex degree is d . Then

$$\chi(G) \leq \mathbf{d + 1}$$

- Intuition
 - If the maximum vertex degree of a graph is low, we can determine a tight upper bound for the chromatic number.
 - However, for graphs with a high maximum degree, this theorem becomes less effective or loses its usefulness.

Proof

- Proof by induction on the number of vertices n
- The statement is true for K_1 the simple graph with one vertex
 - $\chi(K_1) = 1$ and $d = 0$.
- We assume that $\chi(G) \leq d + 1$ for all simple graphs H with fewer than or equal n vertices.
- We should show that $\chi(G) \leq d + 1$ for all simple graphs G with $n + 1$ vertices.

Proof

- Let G be a simple graph with $n + 1$ vertices and maximum vertex degree d .
- Let H be a graph obtained from G by removing a vertex v and its incident edges.
- H has at most n vertices and a maximum vertex degree of d or less. By our assumption, we have $\chi(H) \leq d + 1$.
- **By assigning an unassigned color to v , we can create a $(d+1)$ -coloring of G . Therefore, $\chi(G) \leq d + 1$.**
- In the worst-case scenario, if $\deg(v) = d$ and all vertices have unique colors, exactly one unassigned color remains.

THEOREM 3 (Brooks 1941)

*Let G be a connected simple graph whose **maximum vertex degree is d** .
If G is **neither a cycle graph with an odd number of vertices, nor a complete graph**, then :*

$$\chi(G) \leq d.$$

- Intuition
 - The theorem does not work for K_n and C_{2k+1}
 - This theorem provides a stricter upper limit for the chromatic number.

Chromatic Number $\chi(G)$ and Bounds

- To determine the chromatic number $\chi(G)$ of a graph G , find an equal upper and lower bound, which becomes the chromatic number $\chi(G)$.
- Possible upper bounds for $\chi(G)$
 - Total number of vertices in G .
 - Number of colors in an explicit vertex coloring of G .
 - Maximum degree (d) in G plus one (Theorem 2).
 - Maximum degree (d) in G , if G is not an odd cycle or complete graph K_n (Brooks' theorem).
- Possible lower bound for $\chi(G)$
 - Number of vertices in the largest complete subgraph in G .

Coloring Planar Graphs

◦ **THEOREM 4** (Six Colors Theorem for Planar Graphs)

*The vertices of any simple connected planar graph G can be coloured with **six (or fewer) colours**.*

• **Intuition:**

- For planar graphs, it is possible to color them using 6 colors or fewer.
- Even highly complex planar graphs can be colored using a maximum of 6 colors.

Proof

- Proof by induction on the number of vertices n
- The statement is trivially true when $n = 1$
- Assuming simple connected planar graphs with fewer or equal than n vertices can be colored with 6 or fewer colors
- We aim to prove the same for simple connected planar graphs with $n + 1$ vertices.

Proof

- › Let G be a simple graph with $n+1$ vertices
- G contains a vertex v of degree 5 or less (Previous lecture).
- Create graph H by removing vertex v and its incident edges from G .
- By our assumption, the vertices of H can be colored with 6 colors.
- Reintroduce vertex v .
 - An unassigned color is available because v has a degree ≤ 5 and there are 6 available colors.
 - We color v with this unassigned color.
- This results in a 6-coloring of the vertices in G .

THEORME 5 (Five Colors Theorem for Planar Graphs)

The vertices of any simple connected planar graph G can be colored with five (or fewer) colors.

- **Intuition:**

- For planar graphs, it is possible to color them using only 5 colors or fewer.
- Even highly complex planar graphs can be colored using a maximum of 5 colors.

Proof

- Proof by induction on the number of vertices n
- The statement is trivially true when $n = 1$
- Assuming simple connected planar graphs with fewer or equal than n vertices can be colored with 5 or fewer colors
- We aim to prove the same for simple connected planar graphs with $n + 1$ vertices.

Proof

- Let G be a simple graph with $n+1$ vertices
- G contains a vertex v of degree 5 or less (Previous lecture).
- Create graph H by removing vertex v and its incident edges from G
- By our assumption, the vertices of H can be colored with 5 colors.
- If we reintroduce vertex v , three cases appear
 - Case 1: If there is an unassigned color, we are done
 - Case 2: two colors are not linked path
 - Case 3: two colors are connected with a path

Case 2: two colors are not linked path

By switching one color to be the same as another color, we created an unassigned color.

Case 3: two colors are connected with a path

We simply switch two colors without creating an unassigned color.

Case 3: two colors are connected with a path

The planarity prevents the colors from forming a path due to existing path

Another proof of case 3

- The graph is contracted by removing two edges, resulting in a planar graph with fewer than n vertices.
- The contracted graph is 5-colorable.
- The two edges are reinstated, assigning the original color of v to both v_1 and v_3 .
- Assign to v an unassigned color in the contracted graph
- A 5-coloring of the graph G is achieved by coloring.

THEORME 6 (Four Colour Theorem for Planar Graphs)

The vertices of any simple connected planar graph can be colored with four (or fewer) colours.

Intuition

- It states that only 4 colors are needed to color the regions of any map.
- Conjectured in the 19th century.
- Proven in 1976 with computer assistance.
- It demonstrates the inherent simplicity within complex spatial configurations.

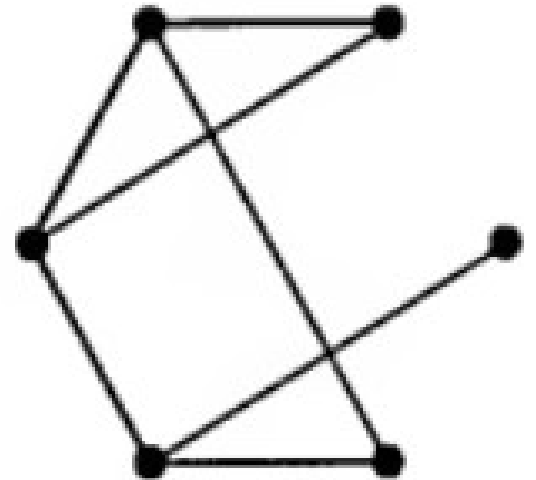
Greedy Algorithm for Vertex Colouring

- Start with graph G and a list of colors 1, 2, 3, ...
- **Step 1:**
 - Label the vertices as $a, b, c, \dots$ in any manner.
- **Step 2:**
 - Identify the uncolored vertex labeled with the earliest letter in the alphabet.
 - Color this vertex with the first color from the list that is not used by any adjacent colored vertex.
- Repeat **Step 2** until all vertices are colored.
- Stop. A vertex coloring of G has been obtained.
- The number of colors used depends on the labeling chosen for the vertices in **Step 1**.

Example

Find a vertex colouring of the following graph G .

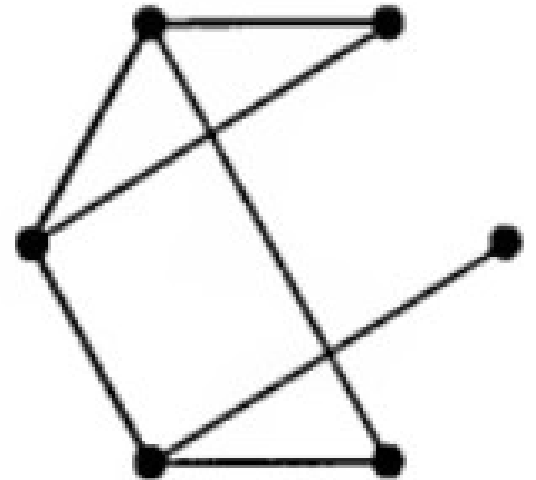
- **Step 1:**
 - Label the vertices as $a, b, c, \dots$ in any manner.
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Example

Find a vertex colouring of the following graph G .

- **Step 1:**
 - Label the vertices as $a, b, c, \dots$ in any manner.
- **Step 2:**
 - Identify the uncolored vertex labeled with the earliest letter in the alphabet.
 - Color this vertex with the first color from the list that is not used by any adjacent colored vertex.



THEORME 6

For any graph G , there is a labelling of the vertices for which the greedy algorithm yields a vertex colouring with $\chi(G)$ colours.

- Proof sketch
 - Take any vertex coloring of G with $\chi(G)$ colors, denoted by 1, 2, 3, ...
 - Sequentially label the vertices colored 1 as a, b, c, ...
 - Label the vertices colored 2 starting from the next available label after the last label used for color 1.
 - Continue this labeling pattern for the vertices colored 3, 4, and so on.
 - The greedy algorithm assigns colors 1, 2, 3, ... in order.
 - As a result, only $\chi(G)$ colors are needed for this labeling.

Coloring Problems: Storing Chemicals

- Certain chemicals react violently when they are in contact.
- The manufacturer plans to divide the warehouse into regions to separate dangerous chemical pairs.
- The dangerous pairs of chemicals are marked with an asterisk in a table.

The objective is to find the minimum number of areas required to safely store the chemicals.

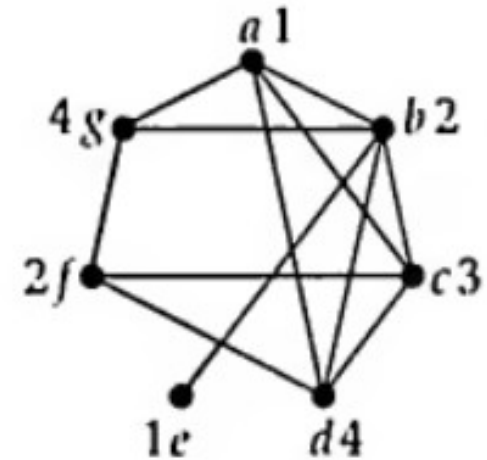
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>a</i>	—	*	*	*	—	—	*
<i>b</i>	*	—	*	*	*	—	*
<i>c</i>	*	*	—	*	—	*	—
<i>d</i>	*	*	*	—	—	*	—
<i>e</i>	—	*	—	—	—	—	—
<i>f</i>	—	—	*	*	—	—	*
<i>g</i>	*	*	—	—	—	*	—

Coloring Problems

- To determine the minimum number of areas required for safe chemical storage, a graph was created.
 - Each vertex represent chemical.
 - Vertices are adjacent when the corresponding chemicals need to be separated.

Assigning chemicals to areas is a vertex coloring problem, with colors corresponding to the areas.

- A vertex coloring leads to a vertex decomposition of the graph, where **no adjacent vertices** are in the same subset.
- The subsets $\{a, e\}$, $\{b, j\}$, $\{c\}$, $\{d, g\}$, representing the chemicals in the four areas.
- The minimum number of subsets required for this problem is the **chromatic number** $\chi(G)$ of this graph.



Coloring Problems: Map Colouring

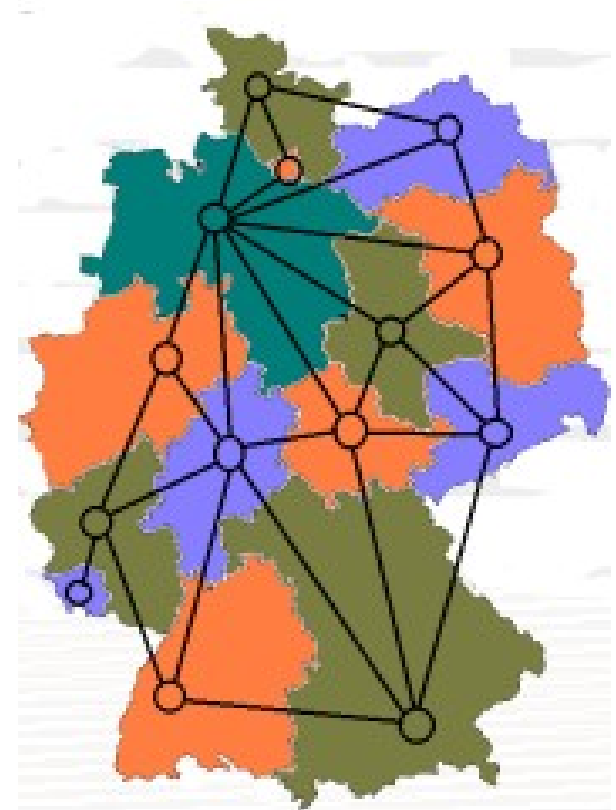
- In 1852, Francis Guthrie proposed the four-color problem:
 - Can all maps be colored with four colors so that neighboring countries have different colors?
- Mathematicians, including De Morgan, Cayley, and Kempe, studied the problem.
- In 1976, Appel and Haken provided a proof using nearly 2000 country configurations and extensive computer analysis.

Even today, no "simple" proof has been discovered for the four-color problem.

Coloring Problems: Map Colouring

- Objective:
 - Assign colors to countries on a map, ensuring adjacent countries have different colors.
- Representation:
 - Use the geometrical dual to represent countries as vertices in a graph, with edges connecting adjacent countries.

Determine the minimum number of colors needed to avoid adjacent countries sharing the same color.



Coloring Problems: Map Colouring

A map is planar graph

- **Theorem 4:** Any map can be colored with 6 colors .
- **Theorem 5:** Any map can be colored with 5 colors .
- **Theorem 6:** Any map can be colored with 4 colors .

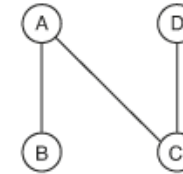
Coloring Problems: Register allocation

- Register allocation can be seen as a graph coloring problem.
- Nodes in the graph represent the live ranges of variables.
- Edges indicate connections between two live ranges.
- The objective is to assign colors to nodes such that adjacent nodes have different colors.

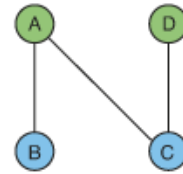
Code Sequence

```
A = ...  
B = ...  
... B ...  
C = ...  
... A ...  
D = ...  
... D ...  
... C ...
```

Interference Graph



Colored Graph



Final Allocation

```
R1 = ...  
R2 = ...  
... R2 ...  
R2 = ...  
... R1 ...  
R1 = ...  
... R1 ...  
... R2 ...
```

The chromatic number represents the minimum number of registers required.

Let's play Sudoku

- Objective:
 - Complete a 9x9 matrix.
- Criteria:
 - In each row, column, and marked 3x3 square, the numbers 1 to 9 should occur exactly once.

Can we model this as coloring problem ?!

			6		3			
	3			1			5	
		9				2		
7			1		6			9
	2						8	
1			4		9			3
		8				1		
	5			9			7	
			7		4			

Conclusion

- Graph coloring is a fundamental concept in graph theory with diverse applications.
- It enables the assignment of colors to vertices by representing entities as vertices and their relationships as edges.
- The goal is to ensure adjacent vertices have different colors, minimizing conflicts.
- Efficient graph coloring algorithms are crucial for optimizing resource allocation and improving system performance.
- Understanding graph coloring techniques helps solve complex optimization problems in various domains.
- Graph coloring has applications beyond computer science, contributing to advancements in diverse fields.

Reference

